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Comments:			
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Task Description:

The interpretation of data from mechanistic studies on radiation biology requires detailed descriptions of the local patterns of energy deposition in the appropriate cellular or sub-cellular sites to explore the effects of radiation quality on biological response. The objective of this project is to provide Monte Carlo (MC) simulations on the local patterns of energy deposition by HZE particles in the appropriate cellular environment including effects of micro-, and macro-geometrical considerations. The calculations focus on interpretation of microscopic “event” or “dose” distributions appropriate to recently initiated studies of HZE radiation induced leukemia in animal and cellular models [NSCOR Program, Colorado State University, R. Ulrich, PI] and those encountered in the production of clustered damage in DNA [e.g., Brookhaven National Laboratory, B Sutherland, PI]. The progenitor cells for leukemia are haematopoietic stem cells that are located in clusters within marrow cavities having dimensions on the order of a few hundred microns. These cavities are surrounded by trabecular bone matrix, which forms a wall of several hundred microns thickness around the marrow. The dimensions and geometry of the bone marrow are too irregular for direct dosimetry measurements and dose-averaging methods are inappropriate for chronic low dose-rate (single particle) exposures. Detailed Monte Carlo charged particle track structure simulation will provide quantitative information on spatial patterns of energy deposition in cellular targets of dimensions as small as a few nanometers on a particle-by-particle basis; these distributions are the starting point to characterize the detailed nature of radiation-induced damage produced by high-LET radiation. The modeling approach will combine the capabilities of full event-by-event Monte Carlo track simulation codes with codes employing condensed history capabilities; the latter is more appropriate for the description of high-energy secondary electrons. The event-by-event ion code PROTRAC (which is part of the biophysical simulation tool PARTRAC) developed at GSF Neuherberg for protons and alpha particles will be extended for heavy charged particle (HZE) interactions using appropriate relativistic theory. Generated secondary electrons that traverse specific regions of interest, e.g., bone marrow cells or their nuclei, will be followed also event-by-event using ETRAC (the electron transport code of PARTRAC) whereas electron transport in adjoining tissues and the surrounding structural material will be followed by the more computer efficient condensed history general purpose MC transport code PENELOPE developed at the University of Barcelona. Appropriate interaction cross sections for a wide range of target materials will be added to the ETRAC and PROTRAC codes, specifically to explore the effects of bone-tissue interfaces on energy deposition in bone marrow cells. Comparison of the spatial pattern of energy deposited in volumes of interest obtained from models of the HZE particles with that obtained from low-LET radiation will provide a firm basis to interpret radiation quality effects on biological response and the mathematical structure developed is amenable to scaling to volumes appropriate to human dimensions for considerations of risk assessment.
The objective of this project is to provide Monte Carlo (MC) simulations on the local patterns of energy deposition by HZE particles in the appropriate cellular environment including effects of micro-, and macro-geometrical considerations. The focus will be on Leukemia, i.e., on the bone/bone-marrow interface from realistic mouse or human dimensions down to single cells.

Little is known on the interaction of HZE particles with the human body. HZE particles are part of the Galactic Cosmic Rays (GCR) which exist in deep space and cannot be shielded completely inside spacecrafts. The latency time of leukemia, 2-3 years, coincides more or less with the mission time for the planned expedition to Mars.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

To summarize the project progress up to now, we have developed a simple, non-bone-site-specific, non-species specific, and computationally-efficient geometrical model for trabecular spongiosa using the general purpose Monte Carlo code PENELOPE. Internal dimensions for our human models are obtained from path-length distributions as measured in different human bones. Details of the model are published in J.A. Gersh et al., Health Physics 93 (2007) 47-59. Our murine model, which we refer to as the scaled mouse model, consists of the human model scaled by a factor of 0.1. Internal dimensions of murine bones published in the literature strongly suggest that simple scaling of the human model suffice. We have extended the track structure transport code PARTRAC to transport HZE particles (bare heavy ions) through liquid water to energies up to 1 GeV/u. We are still working on calculating interaction cross sections and a transport model for metallic calcium, a major component of bone. We have furthermore created a hybrid transport code in combining the geometry with electron and HZE transport in a simple, but not too efficient way. We have performed simulations of an external pencil beam of heavy ions (4He, 16O, 28Si, and 56Fe) in the energy range from 10 to 300 MeV/u. We use the spongiosa deposition factor (SDF) as the dosimetric quantity for interspecies comparisons. The SDF is defined as the ratio of energy deposited in the marrow to the energy deposited in the spongiosa. It also serves for comparisons between different ions used in irradiation of a single species. We find a substantial difference in SDF values between human and mice models for incident energies below 50 MeV/u. This can be attributed to the attenuation of the incident particles, charge particle equilibrium (or the lack of), and the Bragg-peak location. However, for incident energies above 50 MeV/u we see the same SDF values (except for alpha particles) for both, the human and murine models. This strongly suggests that scaling or extrapolating from mouse model findings to human risks should be possible, at least for sufficient fast HZE particles.
Recent progress has been made in the determination of the dielectric properties in calcium and the calculation of interaction cross sections of protons and HZE particles with calcium.While liquid water serves as substitute for soft tissue and bone marrow, we will use metallic calcium, a major component of bone, as substitute for bone in PARTRAC. PENELOPE uses a sum of atomic cross sections due to the atomic constituents (ICRP recommendations) of cortical bone and bone marrow. We are currently updating inelastic proton interaction cross sections with liquid water and have started the derivation of proton interaction cross sections with calcium.

We have derived and adopted a model for the dielectric function (DF) of calcium. It is based on available experimental and theoretical information and uses a modified delta-oscillator extension algorithm. The calculated mean excitation energy is I = 172 eV, which is around 10 % smaller than the ICRU recommended value of 191 ± 8 eV. We are currently deriving the energy loss function with the full dependence on energy and momentum transfer. We are exploring different extension algorithms with the aim of balancing accuracy with applicability. Our current model includes a separation of the energy loss functions into subshells. Parameters are determined using theoretical constraints of the model to eliminate uncertainties. First calculations of mass stopping powers indicate good agreement with ICRU published results. However, the calculation of interaction cross sections requires more detailed information than the averaged calculation of stopping powers. We explore currently different ways to obtain reliable cross sections.

After long deliberations we have adopted the following murine bone model, also called the scaled mouse model. Internal dimensions of murine bones published in the literature strongly suggest that simple scaling of our human model suffice.

Therefore, after longer deliberations, we have dopted the following murine bone model (which we refer to as scaled mouse model). Our scaled mouse model consists of the human model scaled by a factor of 0.1. However, a thorough analysis of micro-CT scans of murine trabecula should be performed to obtain realistic internal dimensions and distributions comparable to those used in the development of the human model. We have established collaboration with Dr. Derek Jokisch from Francis Marion University on further developing and testing the human bone model and helping gathering input data for the rodent model, i.e., analyzing micro-CT scans. Dr Jokisch is part of the University of Florida group that has an extensive history in bone modeling in humans. Also, we have established contacts to Dr. Ted Bateman from Clemson University to obtain micro-CT scans of rodent bones. Dr. Bateman is an expert on bone loss and also interested in radiation induced bone loss.

We also made progress in characterizing the energy depositions in the bone/bone-marrow geometry by introducing the Marrow Dose Heterogeneity Factor (MDHF) and the Spongiosa Deposition Factor (SDF). The Marrow Dose Heterogeneity Factor (MDHF) describes the extent by which the microgeometry and tissue inhomogeneity of the spongiosa region influences the patterns of energy deposition by ionizing radiation within the confines of the marrow cavities of the trabecular spongiosa. The MDHF is defined as the ratio of the actual energy deposited in a region to the homogeneous deposition estimate for a region. MDHFs for human and murine models for internal electron emission within the marrow and bone do not show large inhomogeneities, except for kinematic differences.

We have also performed simulations of an external pencil beam of heavy ions (4He, 16O, 28Si, and 56Fe) in the energy range from 10 to 300 MeV/u. We use the spongiosa deposition factor (SDF) as the dosimetric quantity for interspecies comparisons. The SDF is defined as the ratio of energy deposited in the marrow to the energy deposited in the spongiosa. It also serves for comparisons between different ions used in irradiation of a single species. We find a substantial difference in SDF values between human and mice models for incident energies below 50 MeV/u. This can be attributed to the attenuation of the incident particles, charge particle equilibrium (or the lack of), and the Bragg-peak location. However, for incident energies above 50 MeV/u we see the same SDF values (except for alpha particles) for both, the human and murine models. This strongly suggests that scaling or extrapolating from mouse model findings to human risks should be possible, at least for sufficient fast HZE particles.

We proceed in getting our tools (i.e., the Monte Carlo codes, geometry models, etc) together to address our main hypotheses. The geometry model for the bone/bone-marrow interface looks promising and scaling to mouse dimensions reasonable and possible. The development of interaction cross sections needed for the event-by-event Monte Carlo simulations as well as the development and extension of the codes progresses. First simulation runs have been started and the results are promising. However, we have to continue and extend the simulation runs to higher energies and to better statistics. Simulation runs are computationally very expensive and time consuming.

Bibliography Type:	Description: (Last Updated: 11/09/2018)
Abstracts for Journals and Proceedings	Dingfelder M, Gersh, JA, Jorjishvili, IG, Toburen LH. "Patterns of energy deposition by HZE particles in cellular targets." Poster presented at the Annual NASA Human Research Program Investigators' Workshop 2009, League City, TX, February 2-4 2009. Book of Abstracts, NASA Human Research Program Investigators' Workshop 2009, February 2009. , Feb-2009
Abstracts for Journals and Proceedings	Dingfelder M, Travia A, McLawhorn RA, McLawhorn SL, Shinpaugh JL, Toburen LH. "Emisión de electrones de láminas delgadas de metales y materiales biológicos por impacto de protones: Una prueba para códigos de trazas Monte Carlo." Poster presented at the IV Encontro Sul-Americano de Colisões Inelásticas na Matéria, Rio de Janeiro, Brasil, October 29-31 2008. Book of Abstracts, IV Encontro Sul-Americano de Colisões Inelásticas na Matéria, 2008, p. 3. , Oct-2008
Abstracts for Journals and Proceedings	Fernández-Varea JM, Segui S, Dingfelder M. "Secciones eficaces de producción de rayos x Lalpha, Lbeta y Lgamma de W, Re, Au y Pb por impacto de electrones. Comparación entre cálculos DWBA y medidas experimentales." Poster presented at the IV Encontro Sul-Americano de Colisões Inelásticas na Matéria, Rio de Janeiro, Brazil, October 29-31 2008. Book of Abstracts, IV Encontro Sul-Americano de Colisões Inelásticas na Matéria, 2008, p. 23. , Oct-2008
Abstracts for Journals and Proceedings	Dingfelder M. "New developments in track structure simulations." Invited talk presented at the 54th Annual Meeting of the Radiation Research Society, Boston, MA, September 21-25 2008. Book of Abstracts, 54th Annual Meeting of the Radiation Research Society, Boston, MA, September 21-25 2008. p. S601. , Sep-2008
Abstracts for Journals and Proceedings	Gersh JA, Dingfelder M, Toburen LH. "Simulating the transport of heavy charged particles through trabecular spongiosa." Poster presented by J.A. Gersh at the 54th Annual Meeting of the Radiation Research Society, Boston, MA, September 21-25 2008. Book of Abstracts, 54th Annual Meeting of the Radiation Research Society, Boston, MA, September 21-25 2008. PS3743. , Sep-2008
Abstracts for Journals and Proceedings	Dingfelder M, Travia A, McLawhorn RA, McLawhorn SL, Shinpaugh JL, Toburen LH. "Electron emission from thin metal foils after proton impact: A test for Monte Carlo track structure codes." Poster presented at the 54th Annual Meeting of the Radiation Research Society, Boston, MA, September 21-25 2008. Book of Abstracts, 54th Annual Meeting of the Radiation Research Society, Boston, MA, September 21-25 2008. PS3740. , Sep-2008
Abstracts for Journals and Proceedings	Travia A, Dingfelder M. "Monte Carlo simulations of electron emission yields from biological materials and thin metal foils after proton impact." Invited talk presented at the 20th International Conference on the Application of Accelerators in Research and Industry (CAARI), Fort Worth, TX, August 10-15 2008. Book of Abstracts, 20th International Conference on the Application of Accelerators in Research and Industry (CAARI), Fort Worth, TX, August 10-15 2008. THU-RE08-5. , Aug-2008
Abstracts for Journals and Proceedings	Dingfelder M. "Track structure simulations at relativistic energies: new cross sections for HZE particles." Invited talk presented at the 37th COSPAR Scientific Assembly, Montreal, Canada, July 13-20 2008. Book of Abstracts, 37th COSPAR Scientific Assembly, Montreal, Canada, July 13-20 2008. F22-0003-08. , Jul-2008

Abstracts for Journals and Proceedings	Dingfelder M, Gersh JA, Jorjishvili IG, Toburen LH. "Patterns of energy deposition by HZE particles in cellular targets." Poster presented at the 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, PA, June 30 - July 2 2008. Book of Abstracts, 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, PA, June 30 - July 2, 2008. p. 63. , Jun-2008
Abstracts for Journals and Proceedings	Gersh JA, Dingfelder M, Toburen LH. "Simulating the transport of heavy charged particles through trabecular spongiosa." Poster presented by J.A. Gersh at the 19th Annual NASA Space Radiation Investigators' Workshop, June Philadelphia, PA, 30 - July 2, 2008. Book of Abstracts, 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, PA, June 30 - July 2, 2008. p. 101. , Jun-2008
Articles in Peer-reviewed Journals	Dingfelder M, Segui S, Fernandez-Varea JM. "Distorted-wave ionization and x-ray production cross sections of the K shell of Cu and the L shells of Ag, In, and Sn by positron impact." Physical Review A 2008 Jun;77(6):062710. http://dx.doi.org/10.1103/PhysRevA.77.062710 , Jun-2008
Articles in Peer-reviewed Journals	Dingfelder M, Travia A, McLawhorn RA, Shinpaugh JL, Toburen LH. "Electron emission from foils and biological materials after proton impact." Radiation Physics and Chemistry. 2008 Oct-Dec;77(10-12):1213-7. http://dx.doi.org/10.1016/j.radphyschem.2008.05.046 PMID: 19212448 , Oct-2008
Articles in Peer-reviewed Journals	Dingfelder M, Ritchie RH, Turner JE, Friedland W, Paretzke HG, Hamm RN. "Comparisons of calculations with PARTRAC and NOREC: transport of electrons in liquid water." Radiat Res. 2008 May;169(5):584-94. PMID: 18439039 , May-2008
Dissertations and Theses	Gersh JA. "Simulating the transport of heavy charged particles through trabecular spongiosa." Dissertation, East Carolina University, Greenville, NC, June 2008. , Jun-2008